VS Code Java Setup: Step-by-Step Guide

✅ Part A: Install Required Software

1. Install Java Development Kit (JDK)

• Go to: https://www.oracle.com/java/technologies/javase-downloads.html
OR use OpenJDK from: https://adoptium.net/
• Download JDK version 17 or 21 newer.
• Install it on your system.

📌 Verify Installation: Open Command Prompt (Windows) or Terminal (Mac/Linux) and type:

```
java -version
javac -version
```

2. Install Visual Studio Code (VS Code)

• Download from: https://code.visualstudio.com/
• Install it like any normal application.

3. Install Java Extensions in VS Code

• Open VS Code.
• Go to Extensions (Ctrl+Shift+X).
• Search and install these:
✅ Extension Pack for Java by Microsoft
✅ Debugger for Java
✅ Java Test Runner
✅ Java Dependency Viewer

~~4. Set JAVA_HOME Environment Variable (Windows)~~

~~• Open Start Menu, search 'Environment Variables'.~~

~~• Under 'System Variables', click New:~~

~~- Variable name: JAVA_HOME~~

~~- Variable value: (Path to your JDK folder, e.g., C:\Program Files\Java\jdk-17)~~

~~• Find Path under system variables, click Edit → New → Add:~~

```
~~%JAVA_HOME%\bin~~
```

✅ Part B: Create Your First Java Project in VS Code

1. Open VS Code and Create a New Folder

• Go to File → Open Folder → Create/select folder like JavaProjects

2. Create a Java Project

• Open Command Palette (Ctrl+Shift+P)

• Type: Java: Create Java Project

• Choose:

- No Build Tools
- Name your project (e.g., HelloWorld)

3. Understand Folder Structure

Your project folder will look like this:

```
HelloWorld
└── src
    └── App.java
```

4. Write Your Java Code

By default, App.java is created. Open it and replace with:

```
public class App {
    public static void main(String[] args) {
        System.out.println("Hello, Java with VS Code!");
    }
}
```

✅ Part C: Run and Test the Project

1. Run the Program

• Right-click inside App.java → Click Run Java

OR

• Click the ▶️ icon on the top right

You should see the output:

> *Hello, Java with VS Code!*

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2. Use Terminal to Compile & Run (Optional)

You can also use the terminal:

```
cd src
javac App.java
java App
```

✅ Part D: Extra Tips

Enable Auto Save

• File → Auto Save → After Delay

Enable Code Formatting

• Right-click → Format Document (Shift+Alt+F)

Shortcuts to Remember

• Open Terminal: Ctrl + ~

• Run Code: Ctrl + F5

• Open Command Palette: Ctrl + Shift + P

✅ Part E: Troubleshooting

> *Problem: java not found in terminal*
> *Solution: Set JAVA_HOME correctly*
>
> *Problem: No 'Run' button shown*
> *Solution: Ensure Java Extension Pack is installed*
>
> *Problem: Output not printed*
> *Solution: Check if you saved the file and clicked inside the main method*

🧪 Additional Simple Activity

Try modifying your program:

```
public class App {
   public static void main(String[] args) {
      String name = "Ali";
      System.out.println("Hello, " + name + "!");
   }
}
```

Then compile and run again!